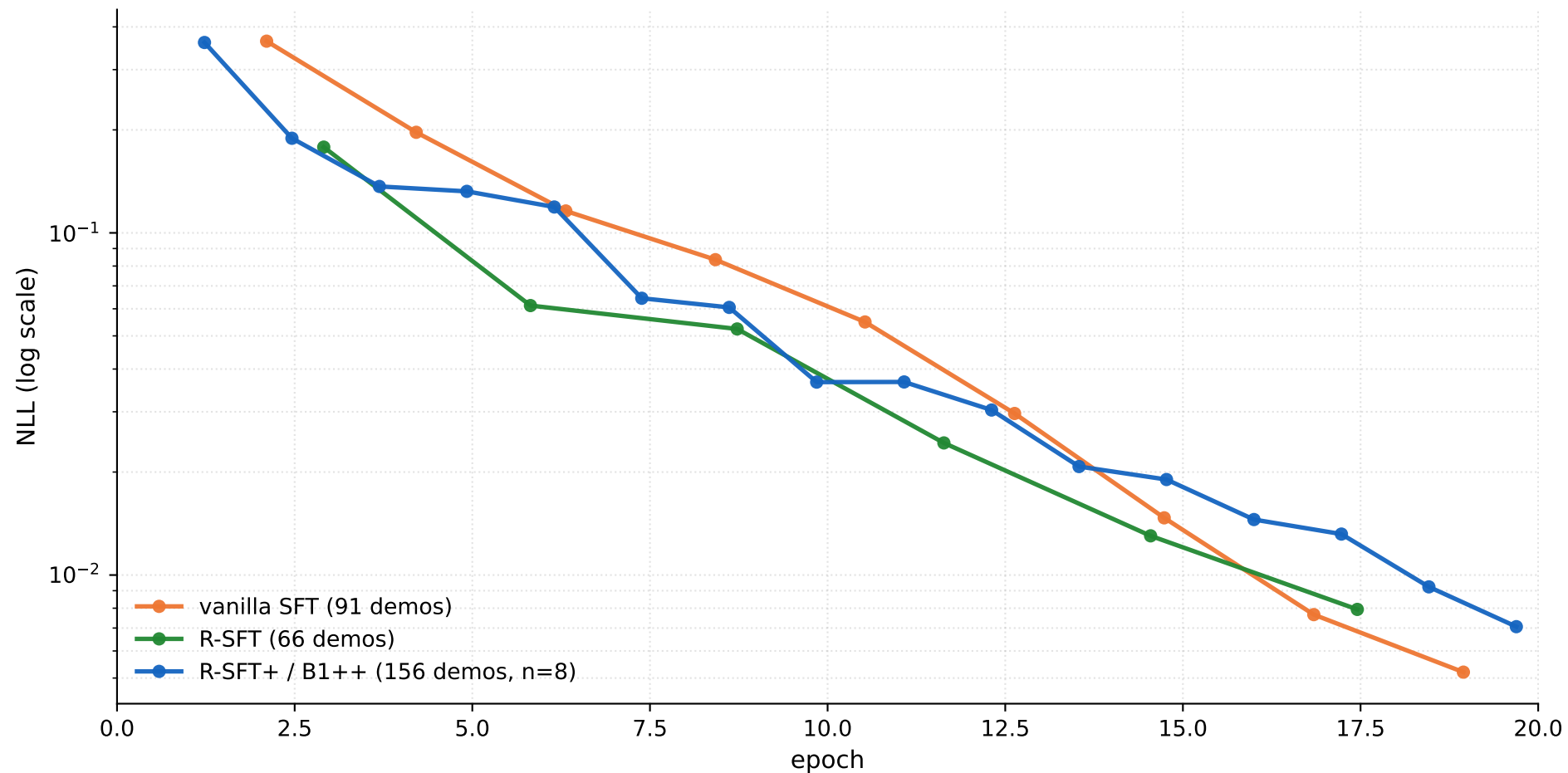


Training NLL — vanilla SFT / R-SFT / R-SFT+ over 20 epochs



Train NLLs collapse to ~ 0.005 - 0.01 by epoch 20 for all three recipes — the model effectively memorizes the SFT targets in NLL terms, regardless of recipe. The downstream LCB-held-out behaviors are still very different (see `sft_progression` / `pareto_frontier`), so the qualitative differences come from what the targets are, not from how well they're fit. With train NLL near zero by epoch 20 and a linear LR schedule decaying to zero, additional epochs would not move these curves meaningfully — the relevant unexplored axes are data (more / harder problems) and LR re-tuning, not more compute on the same data.